CSCI 13500 Software Analysis & Design I

Hunter College City University of New York

LECTURE & RECITATION

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Lecture (02): Tuesday Thursday 9:50 AM - 12:10 PM

Recitation (2R01): Tuesday Thursday 12:30 PM - 2:03 PM

Room: North Building 1001C

COURSE DESCRIPTION

This course is:

1. An introduction to software development, using the C++ programming language. Software development is a skill that involves solving real-world problems by developing computer programs. Breaking a problem down, then creating a series of logical steps (an algorithm) that solves the problem is the conceptual part of the course. The technical component involves translating the solution into a working computer program. A successful student will be able to clearly and logically develop an algorithmic solution to a problem, while being comfortable enough with C++ to transform the algorithm into a working computer program.
2. A preparation for further courses in computer science. This course is one of the "ABCs" of computer science. Students who expect to succeed in more advanced courses in computer science need to go beyond understanding the material presented in this course – they need to master it. It's sort of like the fact that you need to master the alphabet in kindergarten in order to be a successful reader in first grade.
3. An introduction to computers in general. We will not cover: operating systems, networks, databases, etc. This course teaches a specialized skill – developing solutions using computer programming – and only that. You should already possess basic computer skills such as compiling simple computer programs, editing files, manipulating files, etc. If you don't feel comfortable with these sorts of basic computer skills you should come talk to me.
4. An overview of the C++ language. C++ is a huge language with a lot of highly technical details. We will cover the fundamentals of C++, but the focus is on designing algorithms and solving problems.

PRE-REQUISITIES

The prerequisite is CSCI 12700 or instructor's permission. At the very least, you should have written, compiled, and run a program containing iteration while, for and selection if statements.

MAIN COURSE PAGE

Material will largely be posted on the course webpage and/or the brightspace so be sure to check those resources regularly.

TUTORING AND QUESTIONS

Tutoring will be during recitation on Tuesday and Thursday, since there is no TA for this course I highly encourage you to ask questions during lecture and recitation. If you have questions working on your own at home, feel free to email me, and I will try to respond back within a few hours.

GRADESCOPE

You will submit all labs, homework, quizzes, and projects electronically through Gradescope. You will see your grades (including exam grades) on Gradescope as well. An invite email was sent to you before the first lecture. If you have not received it, please let me know as soon as possible.

SOFTWARE

This course is taught in Linux and your programs must be able to run on a Linux platform. On campus, you may use the 1001B lab to do your work for this course. The standard Linux/Unix/Mac OS C++ compiler is g++. If you wish to use a home computer, you can use a Mac. Macs have a Unix command line and g++. You can install Ubuntu Linux. If you want a Linux environment on Windows without installing Linux, follow this excellent tutorial: install Windows Subsystem for Linux (WSL) and Visual Studio in Windows, created by Moududur (Moody) Rahman and improved by Professor Genady Maryash. We have had problems in the past with students programming in a native Windows environment at home, and their programs don't work in the Linux labs and might be incompatible with Gradescope.

OnlineGDB is permitted for temporary use, but has strict limits. Because the site often misinterprets .csv files as source code, you must rename any uploaded data files to .txt to ensure your program runs. Additionally, the site cannot handle large datasets due to a 100 KB per-file limit and overall memory constraints; please use a local environment for more complex assignments.

GRADING

- Projects: 30% (10% each)
- Assignments: 15%
- Quizzes: 20%
- Lecture quizzes: 5%
- Final Exam: 30%

Your worst two quiz grades will be dropped at the end of the semester. Lecture quizzes, projects, and assignments do not get dropped.

LATE AND MAKE UP POLICY

There will be no make ups for any of the assignments or quizzes. You will be given ample time for the assignments and projects and due to the timeframe of the course it is expected of you to begin working on them as soon as possible.

LEARNING OUTCOMES

Show a deep practical knowledge of one widely used programming language. Implement a complete correct program utilizing all basic C++ concepts. Analyze and solve a non-trivial problem by designing and implementing a C++ program on your own. Apply principles of design and analysis in creating substantial programs and projects of realistic scope.

CLASS EXPECTATIONS

1. From the beginning, you will be expected to work independently outside of the lectures. Get started right away – especially if you are going to install your own compiler and/or operating system. The first programming project is assigned during the first week.
2. Absorbing the material and doing the assignments may be challenging. In order to grasp software development concepts one must sit in front of a computer many hours a week actually writing and debugging programs outside of class time. When it comes to programming, the learning is in the doing. There is no substitute for trying and failing, trying and failing, until you finally get your program up and running correctly.

3. Midterm and final exams are largely based on the labs, programming projects, and homework. It is essential that you complete them on time.

POLICY ON ARTIFICIAL INTELLIGENCE AND ONLINE TUTORING

The use of artificial intelligence (AI) or online tutoring websites like chegg.com is strictly prohibited in all exams / quizzes / code reviews. This includes, but is not limited to, the use of AI-generated text, speech, or images, as well as the use of AI tools or software to complete an assignment. Any violations of this policy will result in disciplinary action, up to and including a failing grade for the assignment or course. Our goal is to encourage critical thinking and creativity, and the use of AI detracts from this objective. Students are expected to use their own knowledge, research and analysis to complete coursework.

You can ask AI to give hints or yes / no questions, but not actual codes.

In short, you can use AI as a tutor to ask for questions when working on no-exam-involved assignments, but you are not allowed to submit its codes as your own.

Questions you can ask AI are like:

- Given an input, is the expected output ...?
- Can you show me the next step? Do not give me codes.
- Show me some examples.
- Why is my step wrong?
- What concepts does this question is trying to test?
- Can you give me a similar problem to work?
- Give some ideas or pseudocode to start this problem.

For questions on assignments, you may use https://stackflower.com or https://cplusplus.com/reference/ or https://www.w3schools.com/cpp/.

ACADEMIC INTEGRITY

Hunter College regards acts of academic dishonesty (e.g., plagiarism, cheating on examinations, obtaining unfair advantage, and falsification of records and official documents) as serious offenses against the values of intellectual honesty. The College is committed to enforcing the CUNY Policy on Academic Integrity and will pursue cases of academic dishonesty according to the Hunter College Academic Integrity Procedures.

Please read the academic integrity policy of Hunter College. Here are some examples.

1. In any exam, including but not limited to lab quizzes, midterm, or final, and code reviews, students need to work independently.
2. You are encouraged to discuss non-exam-involved assignments with other students. But after discussion, every student needs to write his/her own codes independently.
3. Copying from another person or from a generative AI system or allowing others to copy work submitted for credit or a grade. This includes uploading work or submitting class assignments or exams to third party platforms and websites beyond those assigned for the class, such as commercial homework aggregators, without the proper authorization of a professor. Any use of generative AI tools must be in line with the usage policy for specific assignments as defined in the course of the syllabus and/or communicated by the course instructor.
4. Using artificial intelligence tools to generate content for assignments or exams, including but not limited to language models or code generators, without written authorization from the instructor.
5. You can never use AI during an exam or a quiz or a code review.
6. In any exam, including but not limited to lab quizzes, midterm, final, or code reviews, all electronic devices should be put in a closed bag.
   a. If a student has any electronic device – including but not limited to computer, calculator, tablet, phone, earbuds – out of a bag, the grade of that student will be zero.
   b. The official time will be displayed in the exam (including lab quiz) environment. Personal watches of any kind – including analog, digital, and smartwatches – are strictly prohibited.

ANTI-HARASSMENT

Bullying, cyberbullying, online hate, intimidation, threats, harassment, and pressure to share schoolwork are all forms of violence. CUNY holds a zero-tolerance stance towards all such acts. The University is committed to prevention of any form of bullying, will respond promptly to threats and/or acts, and will protect victims of bullying from retaliation. As a criminal matter, the New York Attorney General defines cyberbullying as the use of email, websites, instant messaging, chat rooms, text messaging and digital cameras to antagonize and intimidate others. Disrupting a teleconferencing platform (such as Zoom/Skype) is a federal crime.

STUDENTS WITH DISABILITIES AND MEDICAL CONDITIONS

In compliance with the ADA and with Section 504 of the Rehabilitation Act, Hunter College is committed to ensuring educational access and accommodations for all its registered students. Hunter College's students with disabilities and medical conditions are encouraged to register with the Office of AccessABILITY for assistance and accommodation. For information and appointment contact the Office of AccessABILITY located in Room E1214 or call (212) 772-4857 /or VRS (646) 755-3129.

HUNTER COLLEGE POLICY ON SEXUAL MISCONDUCT

In compliance with the CUNY Policy on Sexual Misconduct, Hunter College reaffirms the prohibition of any sexual misconduct, which includes sexual violence, sexual harassment, and gender-based harassment retaliation against students, employees, or visitors, as well as certain intimate relationships. Students who have experienced any form of sexual violence on or off campus (including CUNY-sponsored trips and events) are entitled to the rights outlined in the Bill of Rights for Hunter College.

1. Sexual Violence: Students are strongly encouraged to immediately report the incident by calling 911, contacting NYPD Special Victims Division Hotline (646-610-7272) or their local police precinct, or contacting the College's Public Safety Office (212-772-4444).
2. All Other Forms of Sexual Misconduct: Students are also encouraged to contact the College's Title IX Campus Coordinator, Dean John Rose (jtrose@hunter.cuny.edu or 212-650-3262) and seek complimentary services through the Counseling and Wellness Services Office, Hunter East 1123. Read sexual misconduct CUNY policy.

CHANGE POLICY

Note that details of this document are subject to change if the need arises